MMDU MULLANA OFFICIAL SYLLABUS • BMAT-001 • UNIT 4

Unit 4: Unit IV: Sequences & Series (05 Hrs) and Fourier Series (05 Hrs)

Subject: Mathematics-I (BMAT-001) • **Exam Weightage:** 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Study Time:** 40 mins

Complete Unit IV coverage per MMDU BMAT-001 syllabus: Sequences and series (05 hours): Convergence of sequence and series; Tests for convergence: Comparison test, Ratio Test, Raabe's Test, Logarithmic Test, Cauchy root test, Alternating Series; Leibnitz's rule, conditionally convergent, absolute convergence. Fourier series (05 hours): Euler's formulae, Conditions for Fourier expansions, Fourier expansions of functions having point of discontinuity, change of interval, Odd & Even functions, Half-range series.

1. Convergence Tests: Comparison, Ratio, Raabe's & Logarithmic

1. In-Depth Theory of Infinite Series Convergence & Divergence

Let $\sum_{n=1}^{\infty} u_n = u_1 + u_2 + \cdots + u_n + \ldots$ be an infinite series with partial sums $S_k = \sum_{n=1}^{k} u_n$.

- **Necessary Condition (n^{th} Term Test):** If $\sum u_n$ converges, then $\lim_{\mathbf{n}\to\infty} \mathbf{u_n} = \mathbf{0}$. If $\lim_{n\to\infty} u_n \neq 0$, the series **strictly diverges**.

- **Geometric Series $\sum r^n$:** Converges iff $|r| < 1$.

- **Harmonic p-Series $\sum_{n=1}^{\infty} \frac{1}{n^p}$:** Converges strictly if $\mathbf{p > 1}$, and diverges if $\mathbf{p \leq 1}$.

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2. Systematic Positive-Term Convergence Tests Pipeline

1. **Limit Comparison Test:** If $\lim_{n\to\infty} \frac{u_n}{v_n} = l$ $(0 < l < \infty$, finite and non-zero), then $\sum u_n$ and $\sum v_n$ converge or diverge together.

2. **D'Alembert's Ratio Test:** Let $\mathbf{L} = \lim_{\mathbf{n}\to\infty} \frac{\mathbf{u_n}}{\mathbf{u_{n+1}}}$:

- $\mathbf{L > 1} \implies$ **Convergent** | $\mathbf{L < 1} \implies$ **Divergent** | $\mathbf{L = 1} \implies$ **Test Fails** (use Raabe's or Logarithmic test).

3. **Raabe's Test (when $L = 1$):** Evaluate $\mathbf{R} = \lim_{\mathbf{n}\to\infty} \mathbf{n}\left(\frac{\mathbf{u_n}}{\mathbf{u_{n+1}}} - \mathbf{1}\right)$:

- $\mathbf{R > 1} \implies$ **Convergent** | $\mathbf{R < 1} \implies$ **Divergent** | $\mathbf{R = 1} \implies$ **Test Fails**.

4. **Logarithmic Test (when $L = 1$ with exponents/e):** Evaluate $\mathbf{L'} = \lim_{\mathbf{n}\to\infty} \mathbf{n} \ln\left(\frac{\mathbf{u_n}}{\mathbf{u_{n+1}}}\right)$:

- $\mathbf{L' > 1} \implies$ **Convergent** | $\mathbf{L' < 1} \implies$ **Divergent**.

University Question (6 Marks - Classic MMDU Exam Problem):
Examine the convergence of the series:

$$\frac{1}{2}+\frac{1\cdot 3}{2\cdot 4}+\frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}+\cdots+\frac{1\cdot 3\cdot 5\ldots(2n-1)}{2\cdot 4\cdot 6\ldots(2n)}+\ldots$$

Step 1 🚀: Ratio Test Setup

Here $\frac{u_n}{u_{n+1}}=\frac{2n+2}{2n+1} \implies L=\lim_{n\to\infty}\frac{2n+2}{2n+1}=1 \implies$ **Ratio Test Fails!**

Step 2 ⚡: Apply Raabe's Test

$$R=\lim_{n\to\infty} n\left(\frac{2n+2}{2n+1}-1\right)=\lim_{n\to\infty}\frac{n}{2n+1}=\frac{1}{2}<1$$

🎯 Ratio test fails ($L=1$); Raabe's Test yields $R=\frac{1}{2}<1 \implies$ Series is **Divergent**.

2. Cauchy's Root Test, Alternating Series & Leibnitz's Rule

1. Cauchy's Root Test, Alternating Series & Leibnitz's Rule

A. Cauchy's n^{th} Root Test:

✅ **FUNDAMENTAL LAW / RULE:**

Theorem: Let $\sum u_n$ be a positive-term series and evaluate $\mathbf{L}=\lim_{\mathbf{n}\to\infty}(\mathbf{u_n})^{\mathbf{1/n}}$:

- $\mathbf{L<1} \implies$ **Convergent** | $\mathbf{L>1} \implies$ **Divergent** | $\mathbf{L=1} \implies$ **Test Fails**.

B. Alternating Series & Leibnitz's Rule:

✅ **FUNDAMENTAL LAW / RULE:**

Theorem: An alternating series $\sum_{n=1}^{\infty}(-1)^{n-1}u_n=u_1-u_2+u_3-u_4+\ldots$ $(u_n>0)$ converges if:

1. **Monotonic Decrease:** $u_{n+1}\le u_n$ for all n.

2. **Vanishing Limit:** $\lim_{n\to\infty}u_n=0$.

C. Absolute vs Conditional Convergence:

- **Absolutely Convergent:** $\sum |u_n|$ converges (e.g., $\sum \frac{(-1)^{n-1}}{n^2}$).
- **Conditionally Convergent:** $\sum u_n$ converges via Leibnitz's test, but $\sum |u_n|$ diverges (e.g., $\sum \frac{(-1)^{n-1}}{n}$ and $\sum \frac{(-1)^{n-1}}{\sqrt{n}}$).

University Question (4 Marks - MMDU Question Paper):
Test the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ for absolute and conditional convergence.

Step 1 🚀: Verify Leibnitz's Conditions
Here $u_n = \frac{1}{\sqrt{n}} > 0$. Since $\sqrt{n+1} > \sqrt{n} \implies u_{n+1} < u_n$, and $\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$, the series **converges** by Leibnitz's Rule.

Step 2 ⚡: Check Absolute Series $\sum |u_n|$

$$\sum_{n=1}^{\infty} |u_n| = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$$

Since $p = \frac{1}{2} \leq 1$, $\sum |u_n|$ **diverges**. Therefore, the series is **Conditionally Convergent**.

🎯 $\sum \frac{(-1)^{n-1}}{\sqrt{n}}$ is **Conditionally Convergent** | $\sum \frac{(-1)^{n-1}}{n^2}$ is **Absolutely Convergent**.

3. Fourier Series: Euler's Formulae & Discontinuity

1. Fourier Series: Foundations & Euler's Formulas

Fourier Representation in $(-\pi, \pi)$ or $(0, 2\pi)$:

$$\mathbf{f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]}$$

$$\mathbf{a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\,dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\sin(nx)\,dx}$$

Dirichlet's Conditions & Jump Discontinuity:

At any finite jump discontinuity $x = c$, the Fourier series converges to the average of the left and right limits:

$$\mathbf{S(c) = \frac{f(c^+) + f(c^-)}{2}}$$

University Question (6 Marks - MMDU Question Paper):
Obtain the Fourier series for $f(x) = x^2$ in $(-\pi, \pi)$. Hence deduce $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Step 1 🚀: Symmetry & Coefficients
Since $f(-x) = (-x)^2 = x^2 = f(x)$, $f(x)$ is **even** $\implies b_n = 0$.

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x^2\,dx = \frac{2\pi^2}{3}, \quad a_n = \frac{2}{\pi}\int_0^{\pi} x^2\cos(nx)\,dx = \frac{4(-1)^n}{n^2}$$

Step 2 ⚡: Fourier Expansion & Deduction at $x = \pi$

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(nx) \implies \text{At } x = \pi: \ \pi^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{1}{n^2} \implies \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}$$

🎯 Fourier Series: $x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(nx)$ | Deduced: $\sum_{\mathbf{n=1}}^{\infty}\frac{\mathbf{1}}{\mathbf{n^2}} = \frac{\pi^2}{6}$.

4. Even/Odd Symmetries, Change of Interval & Half-Range Series

1. Even/Odd Symmetries, Change of Interval $(-l, l)$ & Half-Range Series

Parity / Domain	Fourier Coefficients	Series Expansion
Even Function in $(-l, l)$	$b_n = 0,\ a_0 = \frac{2}{l}\int_0^l f(x)dx,\ a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$
Odd Function in $(-l, l)$	$a_0 = 0, a_n = 0,\ b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$	$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$
Half-Range Cosine in $(0, l)$	$b_n = 0,\ a_0 = \frac{2}{l}\int_0^l f(x)dx,\ a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$	Even periodic extension of period $2l$
Half-Range Sine in $(0, l)$	$a_0 = 0, a_n = 0,\ b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$	Odd periodic extension of period $2l$

University Question (6 Marks - MMDU Question Paper):
Find the Fourier series for the triangular wave $f(x) = l - |x|$ in the arbitrary interval $(-l, l)$. Hence deduce $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$.

Step 1 🚀 : Symmetry Check in $(-l, l)$
Since $f(-x) = l - |-x| = f(x)$, $f(x)$ is **even** $\implies b_n = 0$ and $a_0 = \frac{2}{l}\int_0^l (l-x)dx = l$.

Step 2 ⚡ : Compute a_n **and Deduce at** $x = 0$

$$a_n = \frac{2}{l}\int_0^l (l-x)\cos\left(\frac{n\pi x}{l}\right)dx = \frac{2l}{n^2\pi^2}[1-(-1)^n] = \begin{cases} \frac{4l}{n^2\pi^2}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

At $x = 0$: $l = \frac{l}{2} + \frac{4l}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right) \implies \sum_{n=1,3,5\dots}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{8}$.

🎯 Fourier Series: $\mathbf{l - |x|} = \frac{\mathbf{l}}{2} + \frac{\mathbf{4l}}{\pi^2}\sum_{\mathbf{n=1,3,5\dots}} \frac{\mathbf{1}}{\mathbf{n^2}}\cos\left(\frac{\mathbf{n\pi x}}{\mathbf{l}}\right)$ | Deduced: $\sum_{\mathbf{n=1,3,5\dots}} \frac{\mathbf{1}}{\mathbf{n^2}} = \frac{\pi^2}{8}$.

University Solved Examination Questions (Unit 4)

MMDU Official University Solved Examination Questions (Unit 4)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 10 (1 Mark): Failure of Ratio Test on p-Series

Question:
Why does D'Alembert's Ratio Test fail on the p-series $\sum \frac{1}{n^p}$?

Step-by-Step Solution:
Because $L = \lim_{n\to\infty} \frac{u_n}{u_{n+1}} = \lim_{n\to\infty} (1+\frac{1}{n})^p = 1^p = 1$ for every p, so the Ratio Test provides no conclusion and Raabe's Test must be used.

🎯 **Verified Result:** $L = 1$ for all p, hence Ratio Test fails.

📘 Question 20 (4 Marks): Raabe's Test on Higher-Order Series

Question:

Test the convergence of $1 + \frac{1}{2} \cdot \frac{1}{3} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{1}{5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cdot \frac{1}{7} + \ldots$

Step-by-Step Solution:

Here $\frac{u_n}{u_{n+1}} = \frac{2n+2}{2n+1} \cdot \frac{2n+3}{2n+1} = \frac{4n^2+10n+6}{4n^2+4n+1} \to 1$ (Ratio Test fails).

Applying Raabe's Test:

$$R = \lim_{n \to \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \lim_{n \to \infty} \frac{6n^2 + 5n}{4n^2 + 4n + 1} = \frac{6}{4} = \frac{3}{2} > 1$$

Hence the series is **Convergent**.

🎯 **Verified Result:** Convergent by Raabe's Test ($R = 1.5 > 1$).

📘 Question 24 (6 Marks): Full Fourier Expansion of Discontinuous Square Wave

Question:

Obtain the Fourier series expansion for $f(x) = -k$ on $(-\pi, 0)$ and $f(x) = +k$ on $(0, \pi)$. Hence deduce $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$.

Step-by-Step Solution:

Since $f(x)$ is odd, $a_0 = 0, a_n = 0$, and $b_n = \frac{2}{\pi} \int_0^{\pi} k \sin(nx) dx = \frac{2k}{n\pi}[1 - (-1)^n] = \frac{4k}{n\pi}$ for odd n.

$$f(x) = \frac{4k}{\pi} \left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \ldots \right)$$

Substituting $x = \frac{\pi}{2}$ where $f(\pi/2) = k$ yields $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$.

🎯 **Verified Result:** $f(x) = \frac{4k}{\pi} \sum_{n=1,3,5\ldots} \frac{\sin nx}{n}$ | Gregory-Leibniz series $\frac{\pi}{4}$ proved.